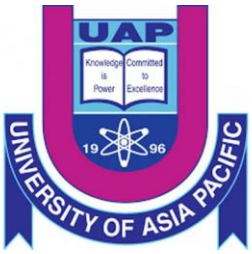

University of Asia Pacific
Department of Computer Science and Engineering
REPORT: 02

Course Code: CSE 402
Course Title: Operating System Lab

Submitted by:

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Semester : 4th Year 1st Semester

Section: C2

Submitted to:

Atia Rahman Orthi,
Lecturer
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Problem Statement:

LAB 2 Worksheet:

Implement FCFS CPU Scheduling for the following scenario where No of processes = 5 and Process informations are-

P id	AT	BT
P0	3	1
P1	5	3
P2	2	2
P3	1	2
P4	6	3

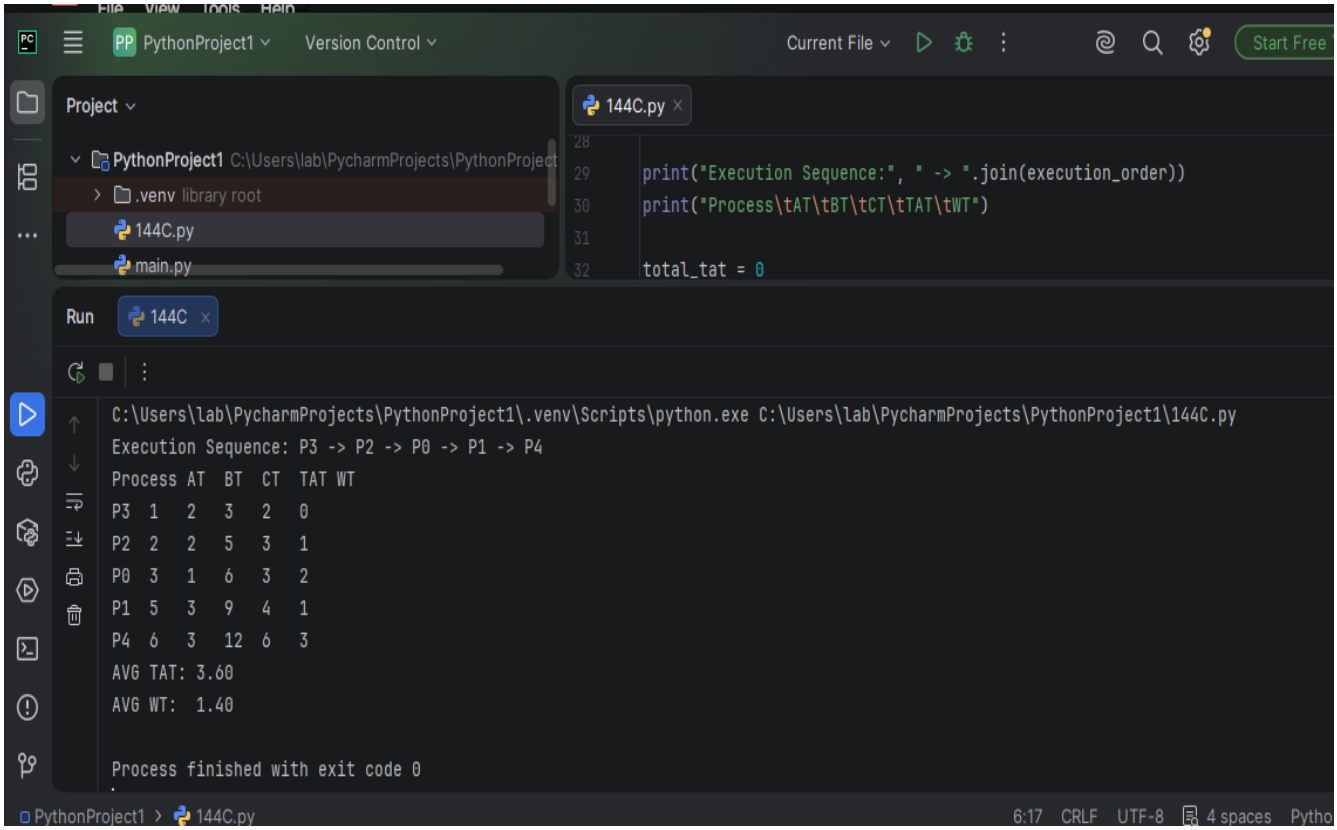
Calculate CT, TAT, WT, AVG TAT and AVG WT. Also represent the process execution sequence.

Problem Solving Approach using Python:

```
processes = [
    ["P0", 3, 1],
    ["P1", 5, 3],
    ["P2", 2, 2],
    ["P3", 1, 2],
    ["P4", 6, 3]
]
processes.sort(key=lambda x: x[1])

current_time = 0
results = []
execution_order = []
```

```
for p in processes:
    p_id, at, bt = p

    if current_time < at:
        current_time = at

    ct = current_time + bt
    tat = ct - at
    wt = tat - bt

    current_time = ct
    execution_order.append(p_id)
    results.append([p_id, at, bt, ct, tat, wt])

print("Execution Sequence:", " ->
".join(execution_order))
print("Process\tAT\tBT\tCT\tTAT\tWT")

total_tat = 0
total_wt = 0

for row in results:
    p_id, at, bt, ct, tat, wt = row
    total_tat += tat
    total_wt += wt
    print(f"{p_id}\t{at}\t{bt}\t{ct}\t{tat}\t{wt}")

n = len(processes)
print(f"AVG TAT: {total_tat / n:.2f}")
print(f"AVG WT:  {total_wt / n:.2f}")
```

Output of the code:



Git repository link:

https://github.com/Rajash144/FCFS-CPU-Scheduling-Lab2

Discussion:

While implementing the FCFS scheduling algorithm, students may face a few common problems. They might forget to sort the processes by arrival time, resulting in an incorrect execution order. Mistakes in calculating CT, TAT, or WT are also common. Another challenge is handling CPU idle time correctly when no process has arrived. Careful implementation and checking each calculation can help avoid these errors.

Conclusion:

The FCFS CPU Scheduling algorithm was successfully implemented using Python. The program correctly determined the execution sequence and calculated CT, TAT, WT, AVG TAT, and AVG WT. This experiment helped in understanding the basic concept and working process of the FCFS scheduling algorithm.